

Highlights

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*Institute of Mathematics and Statistics, Federal University of Rio Grande do Sul
(UFRGS), , , , ,*

Abstract

Keywords:

1. Introduction

Short-term forecasts of the sea state underpin a wide range of offshore and coastal activity, from the planning of installation and offloading windows to the assessment of extreme loads and lifetime fatigue on marine structures ([Orimoloye et al., 2019](#)). Operationally, such forecasts are produced by numerical wave models (NWM), which integrate the wave energy balance equation forward in time given wind forcing, boundary conditions and bathymetry. They are conventionally expressed, distributed and verified in terms of a small number of integrated (bulk) parameters, chiefly the significant wave height H_s and a characteristic period. This convention is deeply embedded in operational practice. Routine intercomparison of wave forecasts has long been established under the World Meteorological Organization (WMO), which now compares eighteen regional and global forecast systems verifying them against in situ buoy observations ([ECMWF, 2026](#)). The fields exchanged for that verification are six integrated parameters: significant wave height, peak and mean zero-crossing periods, mean wave direction and the two components of the 10-m wind.

Nevertheless, the bulk parameters are summaries of a richer quantity, and they are not a sufficient statistic for the purposes to which a forecast is ultimately put. The physically complete description of a sea state is the directional wave spectrum, from which every bulk parameter can in turn be derived, whereas the reverse does not hold ([Holthuijsen, 2007](#)). Because the response of a coastal defence or a marine structure depends on *where in*

frequency the energy is situated, and not only on how much energy is present, two sea states carrying identical total energy — and therefore identical H_s — need not behave alike [Orimoloye et al. \(2019\)](#). The point is sharpest in mixed seas, with double or multi-peaked spectra which no single height-period pair can represent ([Portilla et al., 2009](#); [Violante-Carvalho et al., 2002](#)).

The directional spectrum, although complete, is poorly suited to routine comparison against observations. Operational buoys resolve only its first five directional moments, and existing estimators cannot accommodate more than two directional peaks and may introduce spurious energy ([Gorman, 2018](#)). The frequency spectrum $E(f)$ carries no such ambiguity. It is estimated from the heave record alone, as H_s is, and three of the four wave parameters exchanged for the WMO intercomparison (H_s , T_p and T_z) are derived from it alone. Its absence from routine intercomparison reflects the sampling variability of individual spectral estimates and the absence of a standard spectral skill score in operational practice, rather than a limitation of the observation; where verification has been carried out at the level of spectral components, it has revealed model deficiencies that bulk statistics conceal ([Hanson et al., 2009](#)). $E(f)$ cannot separate wave systems that share a frequency band while travelling in different directions, but it resolves swell and wind sea whenever they are separated in frequency, and it retains the information identified above as lost in the bulk parameters. It is therefore the most complete description of the sea state that a buoy observes without additional assumptions.

Independently of how the forecast is represented, the means of producing it has been changing. In meteorology, data-driven models are now evaluated head-to-head against operational numerical weather prediction systems under shared ground truth and shared metrics ([Rasp et al., 2024](#)), and a first assessment in an operational-like setting reports accuracy comparable to that of a leading operational system for both global metrics and extreme events ([Ben Bouallègue et al., 2024](#)). To our knowledge, comparable gains have been slower to materialise in ocean forecasting [INSERT: reference is likely needed]. The motivation for pursuing them is nonetheless substantial, and is chiefly one of cost: a NWM requires a full forcing chain and considerable computation, whereas a trained model maps recently observed conditions directly to a forecast, and has been shown to reproduce a numerical model’s output at a small fraction of its computational expense ([James et al., 2018](#)), prompting the question of whether such models can act as physical-model surrogates at buoy scale ([Minuzzi and Farina, 2023](#)). By contrast with a NWM, however,

a data-driven model learns the evolution of the sea state (swell dispersion, wind-sea growth and decay) implicitly from historical records rather than from an explicit physical parameterisation, and so obtains its speed at the expense of interpretability and of the physical guarantees a NWM provides by construction.

Within ocean forecasting, data-driven work has concentrated overwhelmingly on the scalar bulk parameters rather than on the spectrum itself. From early artificial neural networks (ANNs) forecasting wave height and period (Makarynsky, 2004), through decomposition-based recurrent approaches to wave-height forecasting (Zhou et al., 2021; Ding et al., 2023), to recent transformer architectures (Kim et al., 2026). Work that does engage with the spectrum generally addresses a different problem. One line estimates a spectrum, or its parametric descriptors, from bulk parameters or from measured platform response, rather than forecasting it forward in time (Naithani and Deo, 2005; Namekar and Deo, 2006; Sakhare and Deo, 2009; Nielsen et al., 2023; Kwon et al., 2026). A second reconstructs or predicts spectra over a spatial domain from atmospheric forcing (Song and Jiang, 2023; Gao and Jiang, 2025; Liu et al., 2025; Cao et al., 2025). Forecasting the spectrum forward in time from a station’s own recent spectral history has received considerably less attention.

Before proposing such a model, a further caution applies. In a comparison across sixteen buoys, Jiang et al. (2024) found that autoregression, gradient boosting and several deep architectures performed almost identically for univariate significant-wave-height forecasting, and concluded that the complex models mostly recover the linear autocorrelation a simple baseline already exploits. We consider this critique sound and it shapes the present study in two ways. First, every forecast is evaluated against a linear and a persistence baseline, rather than assuming the more complex learned model has the advantage. Second, it leaves open whether the same conclusion holds when the target is the spectrum itself rather than a scalar derived from it.

Answering that question is mainly a problem of measurement. When forecast quality is summarised by an error aggregated over the whole spectrum, an error confined to a narrow peak (one or two frequency bins out of several dozen) is diluted by the correctly predicted shoulders and tail. A forecast that merges distinct swell and wind-sea peaks into a single smooth hump can therefore score as well as one that resolves them, or better, and an aggregate criterion will prefer it. The effect has been seen before. A tendency to underestimate the spectral peak was reported for an early neural spectral

estimator and corrected there with an empirical multiplier applied after the fact (Namekar and Deo, 2006), and over-smoothing has been identified as characteristic of data-driven forecasting models in meteorology (Ben Boual-l  gue et al., 2024) and have already been reported in wave forecasting too (Minuzzi and Farina, 2023). The wave-modelling community has long recognised the underlying problem for NWMs. Hanson et al. (2009) note that mean or integral spectral properties give only a general measure of performance and can mask higher-order deficiencies, and for that reason they verify model output within individual wind-sea and swell partitions, an approach more recently applied to long-range spectral forecasts (Rogers and Janiga, 2025). A further complication arises when the models being compared were trained under different objectives. A model optimised for a given error measure tends to score well on that measure, so a comparison based on either model’s training objective cannot show which model better represents the spectrum as a whole.

To our knowledge, no study has yet forecast the wave spectrum autoregressively from a single station’s own spectral history while treating the fidelity of the forecast spectral shape, and not an aggregate error, as the quantity to optimise and report. This paper makes four contributions. (i) We formulate spectral forecasting as sequence-to-sequence regression on the multi-channel spectrum measured at a buoy. The model is a transformer whose input embedding keeps the frequency axis as a structured dimension instead of flattening it, and whose output is parameterised logarithmically, so the recovered spectrum is non-negative by construction and needs no constrained output activation. (ii) We factorise the forecast target into a unit-area spectral shape and a scalar magnitude, forecast the two separately and recombine them at prediction time. This separates the comparatively easy magnitude problem from the harder shape problem. An analogous factorisation is established in short-term load forecasting (Sevlian and Rajagopal, 2018; Guo et al., 2026); here the magnitude is supplied by the physical relation between wave height and spectral energy. (iii) We train against a composite spectral objective combining a distributional divergence (Kullback and Leibler, 1951), an optimal-transport distance (Peyr   and Cuturi, 2019) and a differentiable peak-height term. Each targets a kind of spectral error to which a per-bin criterion is insensitive, and model selection uses a criterion that does not reduce to any one of them. (iv) We evaluate with a peak-resolved diagnostic panel in which spectral peaks are identified by published significance criteria and classified as wind sea or swell (Portilla et al.,

2009), and forecast quality is reported separately for each class, so errors that an aggregate score would dilute stay visible. Each component of (iii) is admitted on evidence through an incremental protocol: a candidate is kept only if it improves the evaluation panel without degrading any other part of it, and only if its measured computational cost justifies the gain.

The remainder of this paper is organised as follows. Section 2 presents the methodology: the problem formulation and the shape/magnitude decomposition of the forecast target (Section 2.1), the buoy record and its preparation (Section 2.2), the model architecture (Section 2.3), the training procedure and hyperparameter search (Sections 2.4 and 2.4.1), the evaluation protocol and its peak-resolved, partition-conditioned diagnostics (Section 2.5), and the incremental ablation strategy under which each component was accepted or rejected (Section 2.6). Section 3 reports the results, and Section 4 discusses their implications and the limitations of the study.

2. Methodology

2.1. Problem formulation

The forecasting task is posed as sequence-to-sequence regression on the time series of wave frequency spectrum measured at a single buoy. At each hourly time step t the record contains five quantities on a fixed grid $f_1, \dots, f_F$ ($F = 47$ bins, 0.0200–0.4850 Hz, non-uniformly spaced with finer resolution at low frequencies): the spectral density $E(f, t)$ [m^2/Hz], the mean and principal wave directions $\alpha_1(f, t)$ and $\alpha_2(f, t)$, and the first and second normalised polar coordinates $r_1(f, t)$ and $r_2(f, t)$. The circular quantities α_1 and α_2 are encoded as (sin, cos) pairs, giving $C = 7$ channels per bin, $\mathbf{x}(f_k, t) = [E, \sin \alpha_1, \cos \alpha_1, \sin \alpha_2, \cos \alpha_2, r_1, r_2](f_k, t)$.

Given the L most recent hourly observations, $\mathbf{X}_t = \mathbf{x}(f_{1:F}, t - L + 1:t)$, the model forecasts a trajectory of τ hourly steps, $\tau \in \{12, 24, 48\}$ h, with a separate model trained and tuned for each horizon. Only the terminal step is treated as the forecast product; the intermediate steps are the autoregressive scaffolding needed to reach it.

Shape/magnitude decomposition of the forecast target. Rather than forecasting the physical spectrum $E(f, t)$ directly, the forecast target is factorised into a shape and a magnitude component, which are modelled separately and recombined at inference:

$$E(f, t) = S(f, t) m_0(t), \quad S(f, t) = \frac{E(f, t)}{m_0(t)}, \quad m_0(t) = \left(\frac{H_s(t)}{4} \right)^2, \quad (1)$$

where S is the unit-area spectrum and $m_n = \int f^n E(f) df$ are the spectral moments. A shape model forecasts $S(f, t + 1:t + \tau)$, a magnitude model forecasts H_s , and the two outputs are multiplied according to Eq. (1). The split separates the comparatively easy magnitude problem from the harder shape problem, and is motivated by the tendency of a single model forecasting $E(f, t)$ directly to underestimate spectral peaks and over-smooth the high-frequency tail. Such a monolithic model is retained as a reference (Section [INSERT: Results section reference]).

Spectral targets are predicted in log space, so the recovered spectrum is non-negative by exponentiation, without an output constraint or clipping. $T_{m02} = \sqrt{m_0/m_2}$, is derived from the forecast spectrum for evaluation while H_s is output directly by the forecast model.

2.2. Data

Source and split. Data are from NDBC station 32012 (Woods Hole Oceanographic Institution Stratus mooring; 19°25'29" S, 85°04'39" W), which reports the five channels of Section 2.1 hourly. The record spans 2016-01-01 00:00 to 2017-12-31 23:00 (17,544 hourly observations, ≈ 2 years). It is split chronologically, without shuffling: 70% for training (to 2017-05-26 15:00), 15% for validation (to 2017-09-13 07:00) and 15% for testing (to 2017-12-31 23:00). The test split plays no part in tuning or model selection, and final results are reported on it. Only temporal generalisation is assessed; generalisation to unseen buoys is not.

Preprocessing and normalisation. All channels are reindexed to a common hourly grid, and gaps are filled by linear interpolation in time (bidirectional at the ends of the record). The directions are interpolated through their sin/cos components and recovered with `atan2`, so a gap is never bridged across the wrong side of the compass. Missing records are few and short. Of the 17,544 hourly records, 124 (0.71%) were absent from the raw files and filled in this way. The five channels share identical timestamps, so each record is either fully observed or fully interpolated. The 124 missing records form 93 gaps, of which 87 last a single hour and only one exceeds 3 h (26 h, 30–31 May 2017, within the validation split). The model therefore sees a strictly regular sequence. Normalisation statistics are fitted on the training split only. The spectral density is scale-normalised, $\tilde{E}(f, t) = E(f, t)/\mu(f)$, with $\mu(f)$ the per-frequency training mean (floored at 10^{-8}); scaling without centring preserves non-negativity, so $H_s = 4\sqrt{m_0}$ can be computed after

Figure 1: Model architecture. Left: the four-stage frequency-structured embedding that maps the $F \times C$ measurement at one time step to a single token. Right: encoder-decoder layout, showing where the growth/decay features and the temporal convolutional front-end enter the encoder, and the autoregressive decoder loop with unit-area renormalisation.

denormalisation. r_1 and r_2 are z -scored, and the sin/cos components are left unscaled.

Targets. Targets are built from the physical density. The shape target $S = E/m_0$ (with m_0 floored at 10^{-12} m^2) is mapped to log space as $y = \log \max(y_{\text{phys}}, \epsilon \bar{y}(f))$, where $\bar{y}(f)$ is the per-frequency training mean of the physical target and $\epsilon = 10^{-3}$. The per-bin floor stays proportional to the energy each bin typically carries and avoids $\log 0$ in calm bins. The same transform is applied to the decoder start token and to every ground-truth decoder input, so the whole autoregressive loop runs in log space. Input/target pairs are formed by sliding a window of L input and τ target steps across each split, giving $T_{\text{split}} - L - \tau + 1$ samples per split.

Growth/decay features. Whether the wave systems in the input window are growing or decaying is informative for their near-term evolution, but it is a property of the sequence of spectra and cannot be read from any single one. Each sample is therefore given twelve scalar features from Dynamic Mode Decomposition (Schmid, 2010). A linear operator A with $\mathbf{x}_{t+1} \approx A \mathbf{x}_t$ is fitted across the L normalised density spectra of the window, restricted to its eight leading singular vectors, and eigendecomposed. Each eigenvalue λ gives a growth rate $\ln |\lambda|$ and an oscillation frequency $\arg \lambda$, both per hour. Because the spectra are real, eigenvalues come in conjugate pairs, and only the member with non-negative frequency is kept; the rank of eight, twice the number of modes retained, leaves enough candidates after this filtering. For the four remaining modes with the largest amplitude (the magnitude of their coefficient for the first spectrum of the window) we keep the growth rate, frequency and amplitude, with zeros where fewer than four modes exist. The features are z -scored on the training split and enter the encoder only.

2.3. Model architecture

The model is an encoder-decoder Transformer (Vaswani et al., 2017) in the pre-normalisation arrangement (Xiong et al., 2020) (Fig. 1).

Frequency-structured embedding. Instead of flattening the $F \times C$ measurement at each time step through one linear layer, the embedding follows its (frequency, channel) structure in four stages:

1. *Per-bin projection.* A linear layer shared across bins maps the C channels at each bin to an 8-dimensional representation (GELU). The channels at one frequency describe the same process, so they are combined before anything is mixed across frequencies.
2. *Frequency identity.* A sinusoidal encoding of each bin’s log-frequency, scaled to the span of $\log f$, is added, together with a learned per-bin residual initialised at zero. Encoding the frequency value instead of the bin index matters because the grid is non-uniform: after a first gap of 0.0125 Hz, the spacing is 0.005 Hz up to 0.0925 Hz, 0.01 Hz from 0.1 to 0.35 Hz and 0.02 Hz from 0.365 Hz upwards.
3. *Frequency-axis convolution.* The dilated convolutional block described below is applied along frequency, with replicate padding, so each bin takes in context from its neighbours. Zero padding would introduce fictitious zero-energy bins beyond the grid edges. The block has the same design as the temporal one but its own weights.
4. *Attention pooling.* A learned query attends over the F bins and pools their 8-dimensional representations into one summary vector, which a linear layer maps to the d_{model} -dimensional token. The weights depend on content, so they can follow the spectral peak, and the parameter count does not depend on F .

Encoder. The twelve growth/decay features are mapped jointly to d_{model} by one linear layer, and the result is added to every encoder token. A fixed sinusoidal positional encoding (Vaswani et al., 2017) is then added; because the sequence is strictly regular, the step index suffices. Before self-attention, the encoder sequence passes through three dilated 1-D convolutions (kernel 3, dilations 1, 2, 4, length-preserving padding), each in a pre-norm residual block with GELU and dropout, so that local trends over a few hours are built into the tokens and attention can concentrate on longer-range dependencies.

Decoder. Decoder inputs pass through the same four-stage embedding, instantiated with $C = 1$, and receive the same positional encoding. The decoder is seeded with the last spectrum of the input window, converted to

the target quantity and log-transformed like the targets. During inference each subsequent input is the model’s previous prediction; during training a causal mask restricts attention to earlier positions. A final linear layer maps each position to a log-spectrum in $\mathbb{R}^F$. At inference, each predicted step is exponentiated, renormalised to unit area (trapezoidal rule on the physical grid) and returned to log space as in the targets before it is fed back, so the autoregressive state remains a valid density. In training, the prediction fed back under scheduled sampling (Section 2.4) is the raw log-space output, so the unit-area constraint is imposed only at inference. The encoder runs once per sample.

Capacity. The model dimension is constructed as $d_{\text{model}} = d_{\text{head}} \times n_{\text{head}}$. For the shape model $d_{\text{head}} = 32$ and $n_{\text{head}} = 8$ are fixed ($d_{\text{model}} = 256$), because they were consistently selected in earlier searches (Supplementary Section S4). The freed search budget goes to the loss weights.

2.4. Training

Loss function. The spectral loss combines three terms evaluated on the log-space prediction $\hat{y}$ and target y ,

$$\mathcal{L} = \lambda_{\text{KL}} D_{\text{KL}} + \lambda_W W_2 + \lambda_P \mathcal{L}_{\text{peak}}. \quad (2)$$

- $D_{\text{KL}}(P_{\text{true}} \parallel P_{\text{pred}})$ is the Kullback–Leibler divergence (Kullback and Leibler, 1951) between the spectra treated as distributions over frequency, $\log P = \log_{\text{softmax}}(y + \log \Delta f)$. Because it is weighted by P_{true} , it charges mainly for mass missing where the true spectrum is concentrated, and a peak that broadens slightly is penalised far less than under a per-bin error.
- W_2 is the 1-D Wasserstein-2 distance between the mass-normalised spectra, computed in closed form in the quantile domain, $W_2^2 = \int_0^1 |F^{-1}(q) - G^{-1}(q)|^2 dq$, without an optimal-transport solver (Peyré and Cuturi, 2019). It tolerates small shifts of a peak but charges a real transport cost for merging two peaks into one, which a pointwise loss does not distinguish from a large displacement.
- $\mathcal{L}_{\text{peak}}$ is a differentiable peak-height term. For each trough-to-trough window k of the *true* spectrum (Section 2.5), the predicted peak height is a softmax-weighted average $\hat{H}_k = \sum_i \hat{E}(f_i) \sigma_k(i)$ with $\sigma_k(i) \propto \exp(\hat{E}(f_i)/\kappa_k)$,

and the loss is the mean of $(\hat{H}_k - H_k^{\text{true}})^2$, where H_k^{true} is the exact window maximum. The temperature $\kappa_k = \max(\kappa_{\min}, c H_k^{\text{true}} (f_{r_k} - f_{l_k}) / (f_F - f_1))$, with $c = 0.15$, scales with the peak’s energy and bandwidth in Hz, so narrow swell peaks get a sharp softmax and broad wind-sea peaks a softer one. Because the windows are fixed by the true spectrum, the term ignores shifts of a peak within its own partition, and so complements W_2 , which measures position but not amplitude. A peak displaced beyond its partition is penalised as missing, as it also is by W_2 .

The weights are tuned per horizon. Implementation details are given in Supplementary Section S2.

The loss ablation (Section 2.6) compares Eq. (2) with a per-bin reference, the frequency-weighted mean squared error in log space, $\mathcal{L}_{\text{base}} = \sum_i w_i (\hat{y}_i - y_i)^2$, with trapezoidal quadrature weights $w_i \propto \Delta f_i$ summing to one. It enters with weight λ_{base} , which is zero in the final configuration (Table 1).

All terms take the log-space quantity the network emits as input, so no denormalisation step enters the gradient path. The per-bin loss weights each bin by its width rather than by the energy it carries, so a log-space error in a near-empty tail bin costs at least as much as the same error at the peak, although it barely affects H_s or T_{m02} . D_{KL} instead weights each bin’s log-space error by the fraction of the true energy in that bin, and $\mathcal{L}_{\text{peak}}$ acts only within the detected peak windows.

Optimisation. We use AdamW (Loshchilov and Hutter, 2019) with a linear learning-rate warmup that rises from 28% to 100% of the sampled rate over the first 5 epochs, so that the first epoch already makes progress, gradient clipping at norm 1.0, and learning-rate halving after 5 epochs without improvement (2-epoch cooldown, suspended during warmup). Without the warmup, the learning rate dominated hyperparameter importance in the search yet correlated only weakly with the objective, with the best and worst trials drawing it from much the same range; this points to unstable early training rather than an optimum the sampler could exploit (Supplementary Section S1). Training uses scheduled sampling (Bengio et al., 2015): at each decoder step the ground-truth previous token is used with probability p , drawn per sample, and the model’s detached prediction otherwise, with p decaying linearly from 1 to 0 over the first 40 epochs. Training runs for at most 100 epochs, with early stopping after 20 epochs without improvement,

and the best-validation checkpoint is kept. Because the validation objective is computed by autoregressive rollout on a short split and is noisy, every decision based on it (scheduling, checkpointing, early stopping and pruning) uses a 5-epoch trailing mean. All settings are listed in Table 1.

2.4.1. Hyperparameter search

Hyperparameters (Table 1) are tuned with the multivariate Tree-structured Parzen Estimator (Bergstra et al., 2011) in Optuna (Akiba et al., 2019), with 18 random startup trials and 80 trials per horizon. From epoch 30, and every 5 epochs thereafter, a median pruner compares a trial’s best smoothed score with the median of completed trials at the same epoch, once at least five trials have completed. It is consulted only when the trial’s own score has not improved on its earlier best for 10 epochs, since trials often dipped below the median around epoch 30, while teacher forcing was still being withdrawn, and then recovered (Supplementary Section S1). The objective (Section 2.5) is computed on the validation split. The global seed is set once before the search and not reset per trial, so that trials differ in initialisation and data order and the sampler can separate hyperparameter effects from run-to-run noise. The selected configuration for each horizon is retrained from scratch with five seeds and evaluated once on the test split; we report mean $\pm$ standard deviation. Training used [INSERT: hardware/GPU specification].

2.5. Evaluation

Baselines. Persistence repeats the last observed spectrum over all τ steps. A stronger baseline is a per-bin linear autoregressive model, rolled out recursively like the Transformer. Each bin is regressed on its own previous p values and an intercept by least squares on the training split, with a ridge penalty of 10^{-6} that only stabilises the solve, since adjacent lags are highly correlated. It works in linear rather than log space on the same target as the Transformer, and for the shape target each predicted step is renormalised to unit area before it is fed back. The order is chosen independently of the Transformer’s input window, from the same candidates $p \in \{12, 24, 48, 96\}$, on the validation split. The selection score is the per-step skill score against persistence, $1 - \text{RMSE}/\text{RMSE}_{\text{pers}}$ with trapezoid-weighted RMSE, averaged over the forecast steps with weights that halve every $\tau/2$ steps. It has no cross-frequency coupling and no nonlinearity, so it isolates the skill that comes from those two sources.

Table 1: Training settings and hyperparameter search space. Log-uniform ranges are marked (log).

<i>Searched</i>	
Input window L	$\{6, 12, 24, 48, 96\}$ h
Batch size	$\{32, 64, 128\}$; $\{32, 64\}$ for $\tau > 24$ h
Learning rate	10^{-3} – 1.5×10^{-2} (log)
Weight decay	10^{-6} – 10^{-2} (log)
Dropout, per-bin representation	$[0.1, 0.3]$
Dropout, d_{model} -wide layers	$[0.0, 0.3]$
Encoder / decoder layers	$\{1, 2, 3, 4\}$ each
$\lambda_{\text{KL}}, \lambda_W, \lambda_P$	$[0.1, 50]$, $[1, 200]$, $[0.01, 20]$ (log)
<i>Fixed</i>	
$d_{\text{head}}, n_{\text{head}}, d_{\text{model}}$	32, 8, 256
Per-bin width	8
Convolutions	kernel 3, dilations 1, 2, 4
λ_{base}	0
Warmup; gradient clip	5 epochs (28–100%); norm 1.0
LR on plateau	factor 0.5, patience 5, cooldown 2
Teacher forcing	p : $1 \rightarrow 0$ over 40 epochs, per-sample draw
Max epochs; early stopping	100; patience 20
Validation smoothing	5-epoch trailing mean
Pruner (epochs)	median from 30, every 5; patience 10
Log floor ϵ	10^{-3}
Peak temperature c ; κ_{min}	0.15; 10^{-4}
Peak windows per spectrum	≤ 4 (lowest frequencies)
DMD modes	4

Core metrics and model selection. We use a peak-fidelity criterion,

$$\text{PF} = \overline{\text{recall}} - \overline{\text{rel.err.}}, \quad (3)$$

the class-averaged peak-separation recall minus the class-averaged peak-height relative error (defined below).

Bulk and shape diagnostics. In physical units and with trapezoidal moment integrals on the true grid, we report:

- RMSE, bias and MAPE of H_s for the recombined spectrum, pooled over all τ forecast steps (MAPE is well behaved here because $H_s \gtrsim 0.8$ m)

throughout the record);

- RMSE and bias of T_{m02} on the final step, which is scale-invariant and can therefore be computed directly on the shape forecast;
- the per-bin scatter index $SI(f) = \text{RMSE}(f)/\overline{E_{\text{true}}(f)}$ for the recombined spectrum, pooled over all τ forecast steps, with the per-bin RMSE.

Peak-resolved, partition-conditioned diagnostics. Because aggregate errors dilute errors confined to narrow peaks (Section [INSERT: Introduction reference]), a set of peak-resolved metrics is computed on the final step. Peaks are detected with the significance criteria of [Portilla et al. \(2009\)](#): a local maximum is rejected if its frequency exceeds 0.4 Hz, its trough-to-trough partition holds less than 5% of the total energy, it has fewer than two bins on either side before the next trough, or it lies between two higher-energy peaks. Each partition is labelled wind sea if $\gamma^* = E_{\text{obs}}(f_p)/E_{\text{PM}}(f_p) > 1$, where E_{PM} is the Pierson–Moskowitz spectrum at the peak frequency; this peak-ratio criterion was proposed by [Portilla et al. \(2009\)](#), building on the spectral-fitting wind-sea identification of [Violante-Carvalho et al. \(2002\)](#). Partitions and labels are taken from the true spectrum and applied to both spectra, so a partition the model misses counts directly as a recall miss. The metrics are:

- the mean number of detected peaks, true and predicted; over-smoothing lowers the predicted count;
- peak-height relative error $|\hat{E}(f_p) - E(f_p)|/E(f_p)$ at each true peak, whether or not the model forms a peak there;
- peak-separation recall, the fraction of true peaks matched by a significant predicted peak within two bins;
- T_{m02} RMSE and bias within each true partition, excluding partitions whose predicted energy is below 5% of the true energy.

The last three are reported overall and separately for wind sea and swell. Wind-sea partitions are broad and fast-evolving, which makes them mainly a magnitude problem; swell partitions are narrow and persistent, which makes them mainly a position problem. A change that helps only one class can disappear in the pooled figure.

The normalisation round trip, the unit-area property of the shape transform and the stated properties of each loss term are checked with unit tests on synthetic spectra (Supplementary Section S5; code in [INSERT: repository]).

2.6. Incremental ablation strategy

Components were introduced one at a time and kept only if they improved the evaluation criteria in use at that stage without degrading others, at a computational cost justified by the gain. Because most candidates interact with capacity and regularisation, a candidate is not inserted into a frozen configuration: the full search of Section 2.4.1 is re-run with it, under the same objective and search space as the comparison run, and the best configurations with and without it are compared. This mixes the candidate’s effect with search noise, so no difference is claimed unless it exceeds the spread across five seeds of the selected configuration.

For the loss terms, where the candidate set is small but interaction with architecture is likely, the design was inverted. Seven arms (the per-bin baseline; each distributional term alone; KL combined with W_2 and with $\mathcal{L}_{\text{peak}}$; and all three) were run at the [INSERT: horizon] horizon with the architecture and training settings fixed, searching only the term weights; the baseline arm was repeated five times to estimate noise. Results are given in Section [INSERT: Results section reference]. The computational cost of each component and the candidates that were not retained are reported in Supplementary Sections S3 and S4.

The protocol has two limitations. Attribution to a single component is only as reliable as the noise estimate, and differences of the order of the seed spread are reported as such. And the peak-resolved diagnostics were introduced partway through, so earlier components were accepted on RMSE-family metrics alone, which can favour over-smoothed forecasts, and have not been re-tested against the full panel.

3. Results

4. Discussion

CRedit authorship contribution statement

Declaration of competing interest

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Data availability

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